

Objective

Design and implement a secure enterprise network that includes VPN connectivity, centralized logging, and attack detection using Splunk. Students will simulate real-world cyber attacks and analyze logs.

Scenario

You are a cybersecurity consultant hired by a mid-sized company with two branch offices. The company requires secure communication between sites, remote access for employees, and centralized monitoring of all activities.

Core Requirements

- Design a network with at least two sites and one remote user
- Implement Site-to-Site VPN between two different vendors (like FortiGate and Palo Alto or Checkpoint)
- Configure Client-Based VPN for remote access
- Create a subnetted IP addressing scheme
- Include LAN, DMZ, and optional management network

Logging & Monitoring

- Send all logs to Splunk
- Include logs from firewalls, servers, and VPNs
- Create at least 1 dashboards

Attack Simulation

Perform at least TWO attacks:

- Brute force attack (SSH or RDP)
- Port scan using Nmap
- Web attack (optional)

Demonstrate detection in Splunk and explain findings.

Infrastructure

- Use virtual machines and/or appliances
- Minimum: 2 firewalls(Different Vendors), 2 servers, 1 attacker machine (Kali Linux)

Deliverables

- Network diagram
- Configuration files (sanitized)
- Screenshots of setup and results
- GitHub or Gitbook repository with full documentation
- Final report (5–10 pages)

Evaluation Criteria

Category	Weight
Network Design	10%
VPN Implementation	30%
Logging & Splunk	20%
Attack Simulation	15%
Documentation (GitHub)	15%
Creativity & Realism	10%

Bonus (Optional) - 15%

- Cloud integration or integration with other instructors project
- Automation scripts
- Advanced Splunk correlation rules

Useful Resources

- Fortinet: <https://opentextbc.ca/fortigatefirewall/>
- Palo Alto: <https://opentextbc.ca/paloalto/>
- Checkpoint: <https://talebi.gitbook.io/checkpoint>